LADR 4e — Alternative Solutions

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This document collects alternative proofs for selected exercises in *Linear Algebra Done Right* (4th edition) by Sheldon Axler. Each solution here differs from the official or standard approach and aims to be more direct or illuminating.

Chapter 6.B

Exercise 1 (6.B.9). *Suppose $e_1, \ldots, e_m$ is the result of applying the Gram–Schmidt procedure to a linearly independent list $v_1, \ldots, v_m \in V$. Prove that $\langle v_k, e_k \rangle > 0$ for each $k = 1, \ldots, m$.*

Solution. *By the Gram–Schmidt procedure, $e_k = \dfrac{f_k}{\|f_k\|}$, so*

$$\langle v_k, e_k \rangle = \frac{\langle v_k, f_k \rangle}{\|f_k\|}.$$

It suffices to show that $\langle v_k, f_k \rangle > 0$.

Substituting the definition

$$f_k = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, f_i \rangle}{\|f_i\|^2} f_i$$

and using $f_i = \|f_i\| e_i$, we obtain

$$\langle v_k, f_k \rangle = \|v_k\|^2 - \sum_{i=1}^{k-1} |\langle v_k, e_i \rangle|^2.$$

By Bessel's inequality,

$$\sum_{i=1}^{k-1} |\langle v_k, e_i \rangle|^2 \le \|v_k\|^2,$$

so $\langle v_k, f_k \rangle \ge 0$.

If equality holds, then $\|v_k\|^2 = \sum_{i=1}^{k-1} |\langle v_k, e_i \rangle|^2$. By Exercise 6.B.3, this implies $v_k \in \operatorname{span}(e_1, \ldots, e_{k-1}) = \operatorname{span}(v_1, \ldots, v_{k-1})$, contradicting the linear independence of $v_1, \ldots, v_m$.

Therefore $\langle v_k, f_k \rangle > 0$, and hence

$$\langle v_k, e_k \rangle = \frac{\langle v_k, f_k \rangle}{\|f_k\|} > 0.$$

Chapter 7.F

Exercise 2 (7.F.3). *Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Prove that*

$$\|Tv\| = \|T\|\,\|v\| \quad \iff \quad T^*Tv = \|T\|^2 v.$$

Solution. *($\Leftarrow$) Suppose $T^*Tv = \|T\|^2 v$. Taking the inner product of both sides with v gives*

$$\langle T^*Tv, v\rangle = \langle \|T\|^2 v, v\rangle.$$

The left-hand side equals $\langle Tv, Tv\rangle = \|Tv\|^2$, and the right-hand side equals $\|T\|^2\|v\|^2$. Hence $\|Tv\|^2 = \|T\|^2\|v\|^2$, and taking square roots yields $\|Tv\| = \|T\|\,\|v\|$.

($\Rightarrow$) Suppose $\|Tv\| = \|T\|\,\|v\|$. Let $S = \|T\|^2 I - T^*T$. We first show that S is a positive operator. For each $u \in V$,

$$\langle Su, u\rangle = \|T\|^2\|u\|^2 - \langle T^*Tu, u\rangle = \|T\|^2\|u\|^2 - \|Tu\|^2 \geq 0,$$

where the inequality follows from $\|Tu\| \leq \|T\|\,\|u\|$. Since T^*T and I are self-adjoint, so is S; thus S is positive. Next, using the hypothesis $\|Tv\| = \|T\|\,\|v\|$,

$$\langle Sv, v\rangle = \|T\|^2\|v\|^2 - \|Tv\|^2 = 0.$$

By 7.43, a positive operator S with $\langle Sv, v\rangle = 0$ satisfies $Sv = 0$. Therefore $\|T\|^2 v - T^*Tv = 0$, that is,

$$T^*Tv = \|T\|^2 v.$$